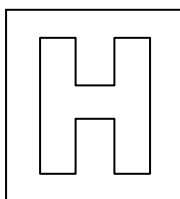


Candidate Name: _____

Class Adm No

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2016 Preliminary Examination II Pre-University 3

H2 Biology

9648/02

Paper 2 Core Paper

16 September 2016

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

This question paper consists of 28 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. Figure 1.1 shows a mid-section of a mammalian DNA molecule, comprising Strand A and Strand B, during the S phase of interphase.

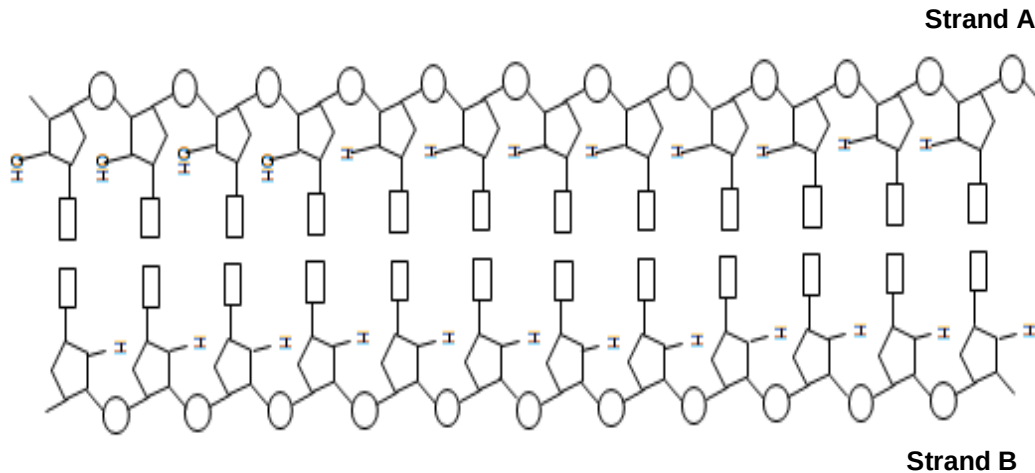


Figure 1.1

- (a) With reference to Figure 1.1,
- (i) account for the difference in the nucleotides in Strand A compared to Strand B.

Strand A has ribonucleotides and deoxyribonucleotides while Strand B has only deoxyribonucleotides;

AND

Strand B is used as the template to guide the synthesis of strand A by semi-conservative replication;

The synthesis of a complementary RNA primer by RNA primase explains the ribonucleotides;

RNA primer is needed as DNA polymerase III can only add deoxyribonucleotides to the free 3' OH end of an already existing polynucleotide;

Max 2

DNA polymerase I removes the RNA primer;

DNA polymerase I replaces RNA primer with complementary deoxyribonucleotides based on the DNA sequence of the intact template strand/Strand B;

Max 1

Figure 1.2 depicts a telomere complex at the end of a chromosome in a human zygote. Telomeres are lengthened during embryonic development to compensate for shortening due to the end-replication problem.

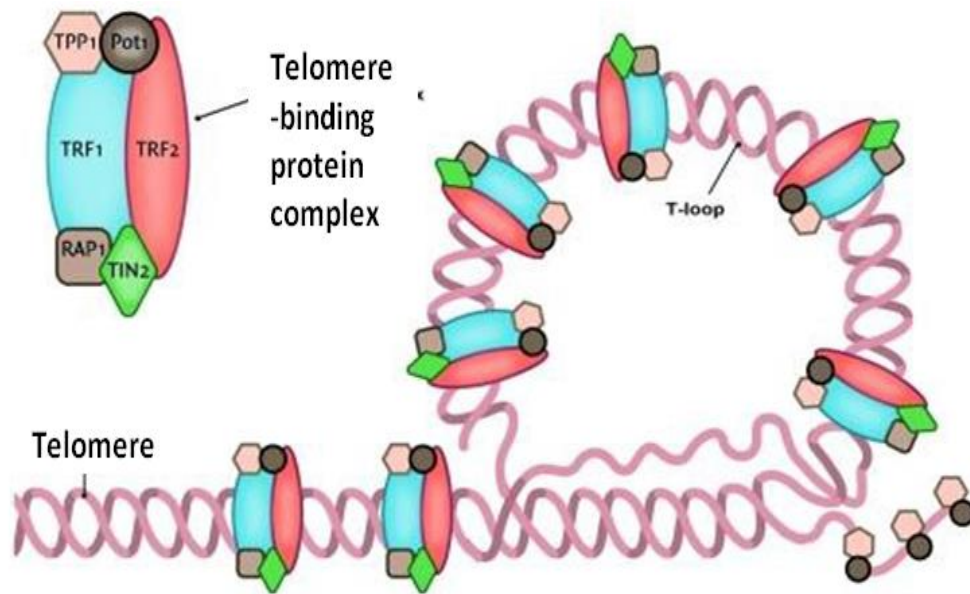


Figure 1.2

(b) Describe two differences between transcription and the process of telomere lengthening.

- .. RNA is used as a template to lengthen the telomere while DNA is used as a template strand in transcription;
 - .. The product synthesised is DNA in the lengthening of the telomere while the product synthesised is RNA in transcription;
 - .. Enzyme involved is telomerase in lengthening of telomere while enzyme involved in transcription is RNA polymerase
 - .. Monomers used in lengthening of the telomere are DNA nucleotides while monomers used in transcription are RNA nucleotides;
- Max 2

A study was conducted to investigate gene expression in yeast. Yeast strain **X** and strain **Y** were used in this study. These strains differed in the location of the *ADE2* gene. In strain **X**, *ADE2* gene is at its normal locus on chromosome 5. An engineered inversion event is responsible for creating strain **Y** where *ADE2* gene locus is near the telomere. Figure 1.3 depicts the loci of the *ADE2* gene in both strain **X** and strain **Y**.

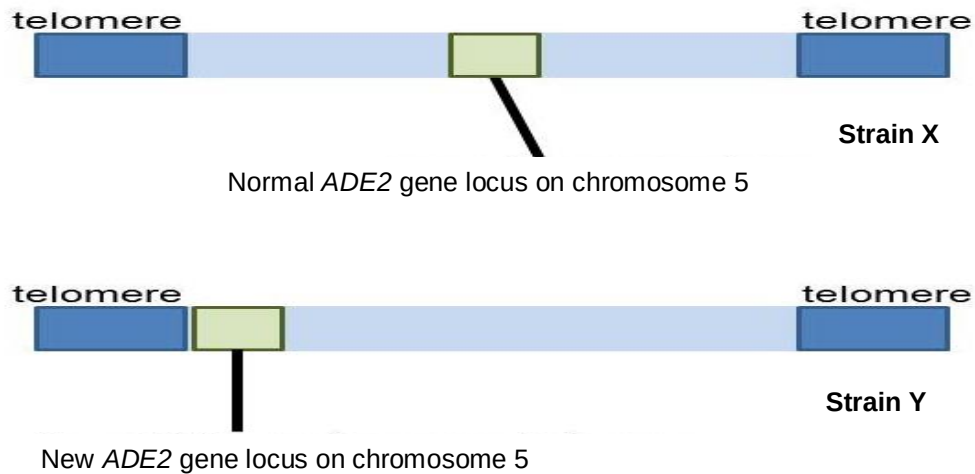


Figure 1.3

The above yeast strains were cultured on specially supplemented agar plates and incubated for one week to produce yeast colonies. Based on the constituents of the Agar, expression of *ADE2* gene prevents accumulation of red pigments and results in cells being white in colour. The colonies produced from this period of incubation are represented in Figure 1.4.

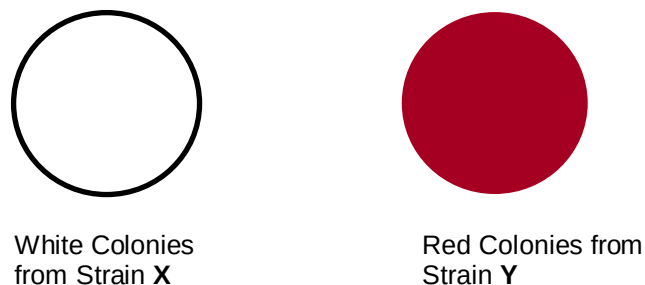


Figure 1.4

Previous studies have shown that some proteins bind to yeast telomeres to recruit histone deacetylase enzymes. These enzymes then act on the chromatin regions around the telomeres.

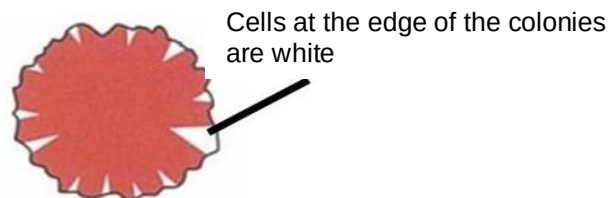
- (c) Using the above information, explain the difference in colony colour between strain X and strain Y after one week of incubation.

ADE2 gene expression is affected by histone deacetylase recruited by telomere bound proteins in Strain Y while the gene is not affected in strain X due to different positions in the genome;

AND

Histone deacetylase remove acetyl group from the lysine tails of the histone proteins;
The histone proteins regain positive charges;
This increases ionic interactions between histone proteins and DNA;
Result in closer packing of DNA around histone proteins;
Prevents transcription factors/RNA polymerase from accessing promoter of ADE2 gene;
 Max 3

The colonies from both strains were allowed incubate for an additional week. Following this, it is then observed that some cells at the edges of the Strain Y colonies appeared white in colour, as depicted in Figure 1.5.



Appearance of colonies from Strain Y after two weeks of incubation

Figure 1.5

- (d) Explain why the cells at the edges of Strain Y colonies appear white in colour after two weeks of incubation.

Telomere length shortens as cells undergo repeated rounds of DNA replication;
Short telomeres in cells at the edge of colony unable to bind to proteins which recruit histone deacetylase;
RNA polymerase and transcription factors are able to bind to ADE2 promoter sequence to form transcription initiation complex;
ADE2 gene is expressed;

[Total: 13]

2. In the cytosol, chaperone proteins can bind reversibly to newly synthesized polypeptides and shield them from other molecules. This prevents the R groups of amino acids from forming bonds with the wrong molecular partners, hence allowing the polypeptides to fold correctly.

Figure 2.1 below shows how chaperone proteins work.

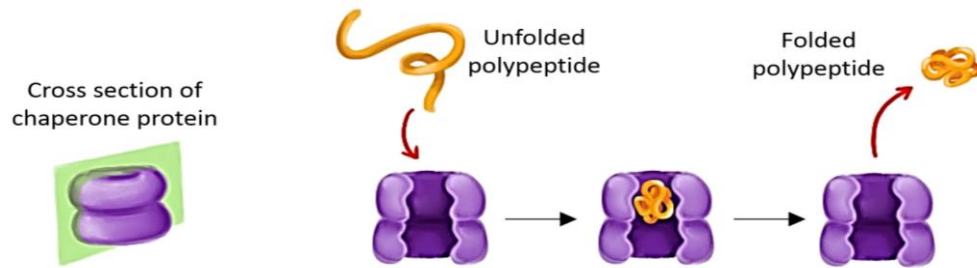


Figure 2.1

- (a) With reference to Figure 2.1, explain how chaperone proteins regulate gene expression.

Chaperone proteins regulate gene expression at the post-translational level;
 Chaperone proteins support production of functional proteins;
 Chaperone proteins provide them with an isolated environment to fold properly;
 Max 2

[2]

- (b) Describe how cytosolic proteins are degraded in cells.

Proteins to be degraded are tagged with ubiquitin molecules;
 Large protein complexes called proteasomes recognise the ubiquitin and degrade the tagged cytosolic proteins;
Proteases and peptidases may also degraded cytosolic proteins;
 Proteins are degraded into smaller peptides or amino acids by hydrolysis of peptide bonds;
 Max 2

[2]

When a protein fails to fold correctly upon synthesis or misfolds at a later stage in its cellular life time, it can no longer fulfil its biological function. Protein misfolding may occur due to mutations in genes or interference with expression of chaperone proteins. Figure 2.2 shows how misfolded proteins may aggregate in a cell.



Figure 2.2

- (c) With reference to Fig. 2.2, suggest why misfolded proteins tend to aggregate in the cell.

Hydrophobic amino acids are exposed in misfolded protein;
Hydrophobic interactions between these amino acids of multiple misfolded proteins cause aggregation;
 Misfolded proteins may not have a shape that is complementary to the active site of proteases for degradation;
 Protein aggregates are too large to enter proteasome to be degraded;
 Max 2;

[2]

An example of a misfolding disease is Bovine spongiform encephalopathy (BSE), a neurodegenerative disease in cattle and commonly referred to as mad cow disease. In BSE, there is an aggregation of misfolded prion protein in the neurons of cattle, which results in degeneration of the brain and spinal cord.

Hereditary forms of BSE have been associated with a mutation in the brain prion protein (PrP) gene, which encodes the PrP protein. The PrP protein is a transmembrane glycoprotein found on the surface of neurons. Figure 2.3 shows the ribbon model of a normal PrP protein.

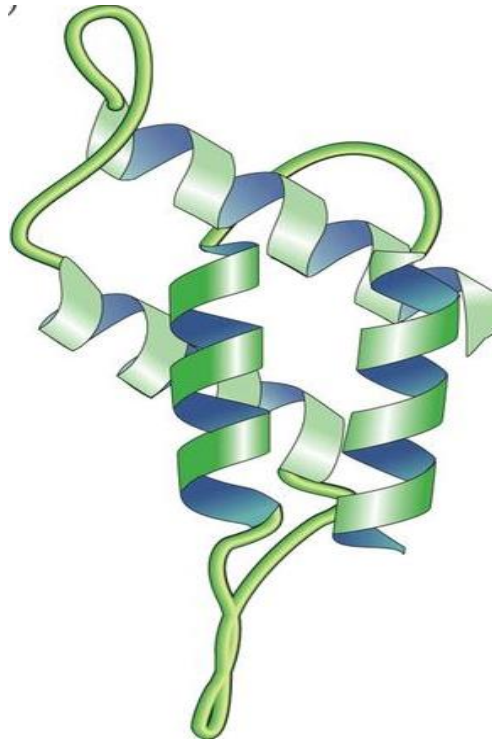


Figure 2.3

- (d) Explain how the PrP protein would maintain its position within the cell membrane.

- ... Polar and charged amino acids of the PrP will interact with the phosphate heads of neighbouring phospholipids by hydrophilic interactions;
 - ... Non-polar amino acids of the PrP will interact with the fatty acid tails of neighbouring phospholipids by hydrophobic interactions;
 - ... Position of PrP is also stabilized by cytoskeletal filaments within the cytoplasm;
- Max 2

- (e) Describe how the PrP protein on the cell surface membrane is formed after its polypeptide chain has been synthesized.

Following synthesis, the PrP polypeptide is inserted into the lumen of the RER;
 The PrP polypeptide chain folds into its 3D conformation in the lumen and undergoes post-translational modification;
Transport vesicles containing PRP buds off from the RER and travel towards and fuses with the cis face of the Golgi apparatus;
 The PrP protein is further modified to become a functional glycoprotein;
 The protein gets embedded in the GA membrane and is packaged into a Golgi vesicle which buds off from the trans face of the Golgi apparatus;

Max 3

AND

The Golgi vesicle with the embedded PrP moves towards and fuses with the cell surface membrane to insert protein into cell membrane;

4]

A human version of BSE called variant Creutzfeldt-Jakob disease (vCJD) is believed to be caused by eating beef products contaminated with central nervous system tissue from infected cattle. Studies have shown that the introduction of misfolded cattle PrP by consuming contaminated meat can cause normal human prion protein to misfold and form aggregated plaques in the brain.

Health advisories have cautioned that cooking contaminated meat will not provide protection from infection that would lead to vCJD.

- (f) Suggest why cooking contaminated meat products will not provide protection from contracting vCJD.

Heat from cooking is unable to denature misfolded PrP;
 PrP is able to refold into infectious form after heating/cooking;
 Max 1

[1]

[3]

3. Burmese cats, *Felis catus*, show discontinuous variation in the colour of their eyes and hearing ability. A large scale investigation was conducted to study the inheritance pattern of eye colour and deafness in such cats.

Pure-breeding blue eyed cats which were deaf were crossed with pure-breeding yellow eyed cats with normal hearing.

The F_1 all had yellow eyes and normal hearing.

Female cats from the F_1 were crossed with male cats with blue eyes and were deaf. 668 offspring were produced.

The observed numbers of F_2 cats with each phenotype were as follows:

Blue eyes, deaf	259
Blue eyes, normal hearing	72
Yellow eyes, deaf	53
Yellow eyes, normal hearing	284

- (a) State what is meant by pure-bred cats for eye colour and hearing ability.

Cats are homozygous for eye colour and hearing ability;
Cats have identical alleles at the loci for eye colour and hearing ability;
Max 1

.....
... [1]

A chi-squared test, with the guide of Table 3.1, was planned to evaluate the difference between observed and expected results.

Table 3.1

Degree of Freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

167 Blue eyes, deafness, 167 Blue eyes, normal hearing, 167 Yellow eyes, deafness, 167 Yellow eyes, and normal hearing;
1 Blue eyes, deafness: 1 Blue eyes, normal hearing: 1 Yellow eyes, deafness: 1 Yellow eyes, normal hearing;
Equal ratio for all 3 phenotypes;
Max 1m

using F_1

..... [1]

- (c) With reference to Table 3.1, outline how the significance of the difference between observed and expected results can be evaluated with 98% confidence (Formulae not required).

Carry out chi-squared test with degree of freedom =3 and $p=0.02$;

If calculated χ^2 is >9.84 difference is significant;

If calculated χ^2 is <9.84 difference is not significant;

If calculated χ^2 is <9.84 difference is due to chance deviation;

Max 2

[2]

The diff
be sign

(d)

Let Y represent the allele coding for yellow eyes.
Let y represent the allele coding for blue eyes.
Let H represent the allele coding for hearing ability.
Let h represent the allele coding for lack of hearing ability.

F1 phenotype: Female cat with yellow eyes and normal hearing X Male cat with blue eyes and deafness

F1 genotype: $\frac{Y}{y} \frac{H}{h}$ X $\frac{y}{y} \frac{h}{h}$

Gametes: $\frac{Y}{+} \frac{H}{+}$, $\frac{y}{+} \frac{h}{+}$, $\frac{Y}{+} \frac{h}{+}$, $\frac{y}{+} \frac{H}{+}$, $\frac{y}{+} \frac{h}{+}$

F2 genotypic ratio: $\frac{Y}{+} \frac{H}{+}$, $\frac{Y}{+} \frac{h}{+}$, $\frac{y}{+} \frac{H}{+}$, $\frac{y}{+} \frac{h}{+}$, $\frac{Y}{y} \frac{H}{h}$, $\frac{Y}{y} \frac{h}{h}$, $\frac{y}{y} \frac{H}{h}$, $\frac{y}{y} \frac{h}{h}$

F2 phenotypic ratio: 284 Yellow eyes, normal hearing : 53 Yellow eyes, deafness : 72 Blue eyes, normal hearing : 259 Blue eyes, deafness

Definition of alleles;

Correct matching of F1 phenotypes and genotypes;

Correct gametes (must be circled) + distinguishing parental and recombinant gametes;

Correct linking of F2 genotype to phenotype;

[4]

The coordinated activity of higher order organisms like cats relies upon a continuous input of information from the internal and external environments. Information is in the form of stimuli, which are detected by receptors. One such receptor is called a Pacinian corpuscle, which is found in the skin.

Fig. 3.1 shows the electrical activity recorded by two microelectrodes inserted into:

- the axon within the Pacinian corpuscle at site **P**, and
- the axon of the sensory neurone leaving the corpuscle at site **Q**

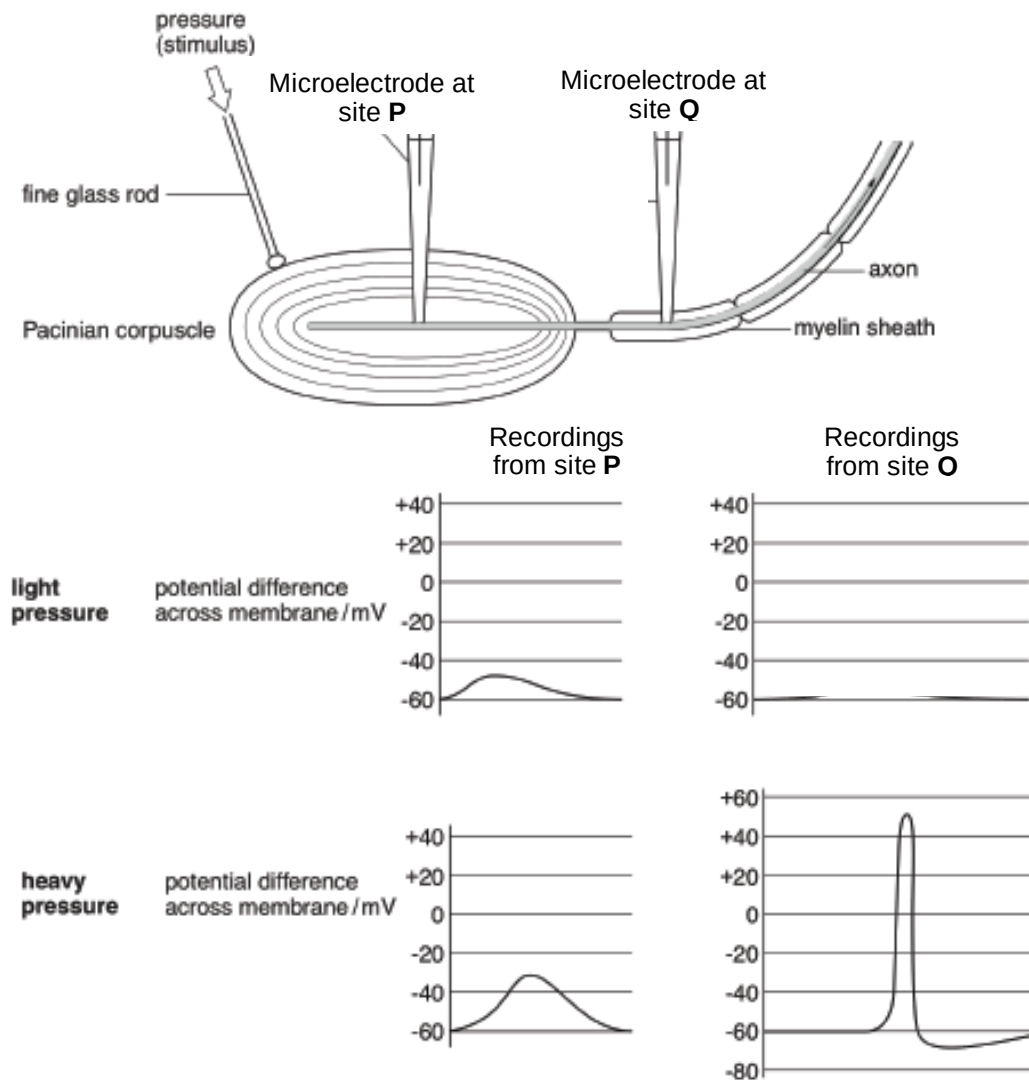


Figure 3.1

- (e) With reference to Figure 3.1, account for the changes in electrical activity in site **P** and **Q** as increasing pressure is applied to the Pacinian corpuscle.

When light pressure is applied, potential difference at **P** increases from -60 to -55mV and there is no change in potential difference at **Q**;

Depolarization at **P** produces a graded potential that is below threshold potential and therefore no Action potential is generated and transmitted to **Q**;

When heavy pressure is applied, potential difference at **P** increases from -60 to -30mV and potential difference at **Q** increases from -60 to +40mV;

Depolarization at **P** produces a potential that is above threshold due to the opening of more voltage-gated sodium channels;

This generates an action potential in line with all or none principle;

Depolarisation of adjacent region of axon from influx of sodium ions at **P** during action potential generates action potential at **Q** results in propagation of impulse from **P** to **Q**;

Max 2m

[4]

[Total: 12]

4. Rhinoviruses are the most common viral infectious agents in humans and are the predominant cause of the common cold. Symptoms include sore throat, runny nasal congestion, sneezing and cough which may be accompanied by muscle aches and loss of appetite.

The structure of the human rhinovirus is depicted in Figure 4.1. Human rhinoviruses are composed of a capsid that contains four viral proteins **VP1**, **VP2**, **VP3** and **VP4**. Each virion has one copy of single-stranded positive sense RNA genome of between 7200 and 8500 nucleotides in length.

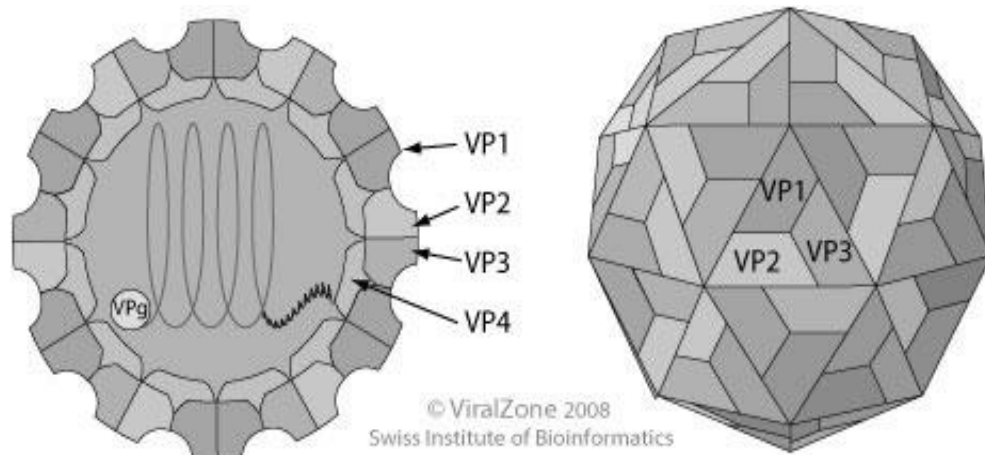


Figure 4.1

Despite structural differences, both human rhinovirus and the influenza virus are able to penetrate respiratory epithelial cells for infection.

- (a) Explain how it is possible for both human rhinovirus and the influenza virus to be able to penetrate human respiratory epithelial cells.

Surface proteins on human rhinovirus and the Influenza virus can bind to different receptors on human respiratory epithelial cells;
Max 1

..... [1]

The enzyme neuraminidase plays an important role after new influenza virions exit their host cells from each round of infection.

- (b) Describe the role of neuraminidase in the reproductive cycle of the influenza

Neuraminidase prevents the clumping of influenza virions to host cell/among themselves after they have budded from cell membrane;
It cleaves the attachment/bond between the HA on influenza and sialic acid residues on the host cell membrane;
Neuraminidase therefore promotes the spread of progeny influenza virions from the host cell surface to infect other host cells;
Max2

- (c) With reference to Figure 4.1, explain why it is unlikely that the newly assembled human rhinovirus exit their host cells in the same way as influenza virions.

Human rhinovirus is a non-enveloped virus;
When influenza viruses bud from host cell, they acquire part of the host cell surface membrane as their viral envelop;
Human rhinovirus is likely to lyse host cell for release;
Max2

[2]

H7N9 is a subtype of Avian Influenza virus, which normally circulates amongst the avian populations with some variants known to occasionally infect humans. An H7N9 virus was first reported to have infected humans in 2013 in China and now this virus has spread among humans. Figure 4.2 shows the proposed evolution of H7N9 virus.

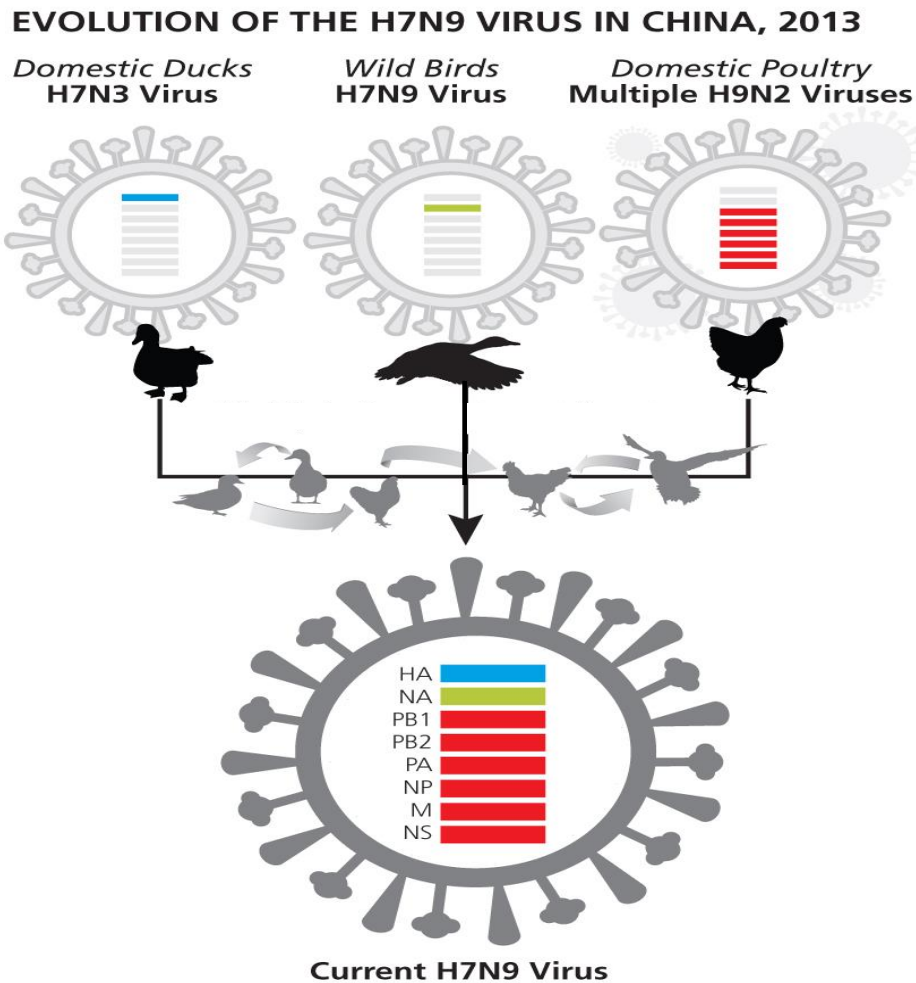


Figure 4.2

(d)

Antigenic shift;
Where H7N3, H7N9 and H9N2 strains infect a series of intermediate host cells at the same time;
The cell is replicating the viral genomes of multiple strains at the same time;
RNA segments are randomly assembled from multiple strains during replication of viral genomes;
Reassortment of genes from these strains occurs resulting in the new combination of RNA segments for H7N9;
Max 3

A

[3]

Today viruses are considered an exception to the cell theory which states that the basic units of life are cells. Viruses have been referred to as “organisms at the edge of life”.

(e) State one characteristic of viruses that may classify them as being

Adapt to unfavourable environments by exhibiting high mutation rates which increase genetic variation;
Able to reproduce by taking over the genetic machinery of its host cells;
Certain viruses are able to respond to external stimuli by switching reproductive cycle;
Contains nucleic acid (DNA or RNA) as its genome to pass on genetic characteristics from one viral generation to the next;
 Max 1

.....
 They are acellular and contain no cytoplasm or cellular organelles;
 Viruses do not grow, divide or increase in size;
 Viruses do not move on its own and can only be carried by host cells or extracellular medium;
 Viruses have no ability of regulating internal environment;
 Viruses carry out no metabolism on their own;
 Viruses lack enzymes to carry out nucleic acid, protein or ATP synthesis.
 Max 1

... [1]

al: 10]

5. A study was carried out to measure the concentrations of glucose and insulin in the blood of students. The results are summarised in Fig. 5.1.

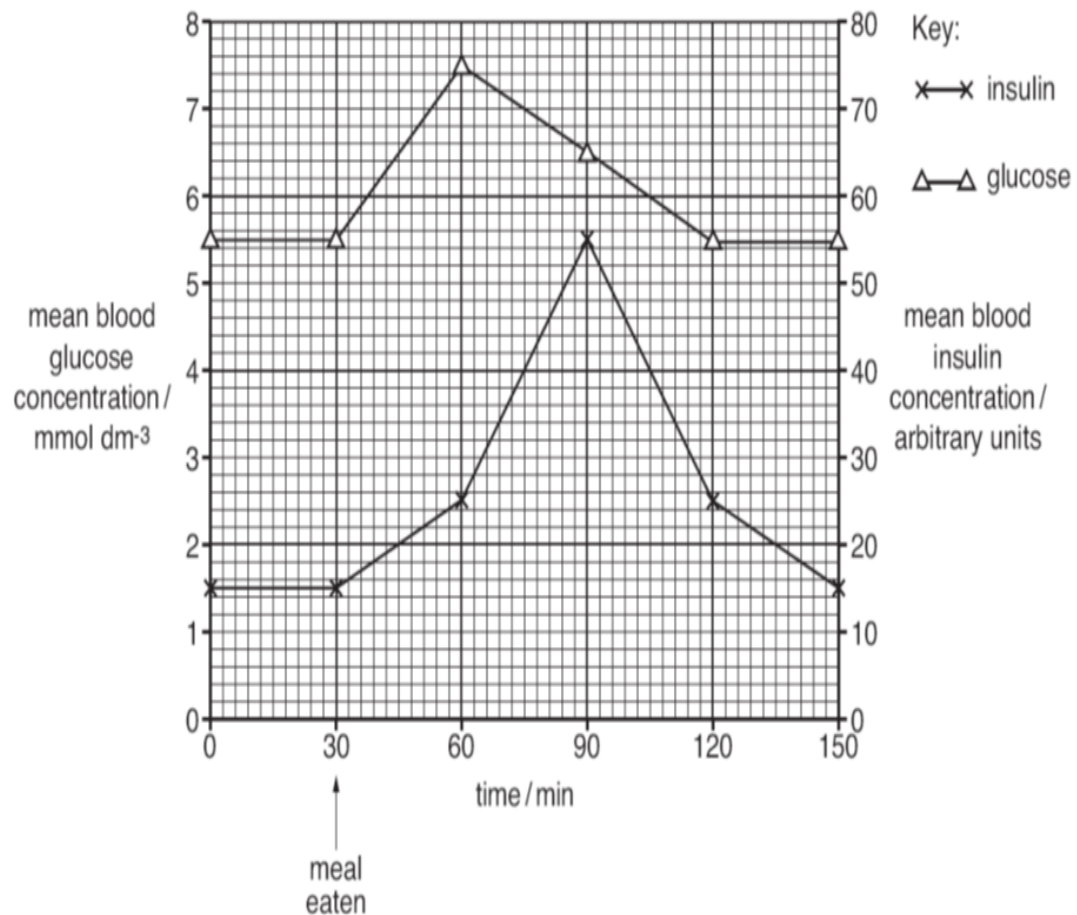


Figure 5.1

- (a) With reference to Figure 5.1, account for the relationship between the concentration of glucose and the concentration of insulin after a meal

Rise in glucose concentration from 5.5 to 7.4 mmol/dm³ between 30 to 60min is detected by β cells islets;

This results in the increase in insulin secretion into the blood from 15 to 55 arbitrary units between 30 to 90min;

AND

Insulin binding to receptors on the liver / muscle/ adipose cells;
Insulin signalling results in

- Increased glucose uptake; and utilization in target cells;
- Increased glycogenesis in liver cells;
- Increased protein synthesis;
- Inhibition of glycogenolysis and gluconeogenesis;

Max 2

This results in glucose concentration being lowered from 7.4 to 5.5mmol/dm³ between 60 to 120min;

Decreased blood glucose levels serves as negative feedback to the β -cells which decreased the secretion of insulin, lowering insulin levels from 55 to 15 arbitrary units from 60 to 150mins;

[5]

Figure 5.2 outlines some steps in glucose metabolism in human muscle cells.

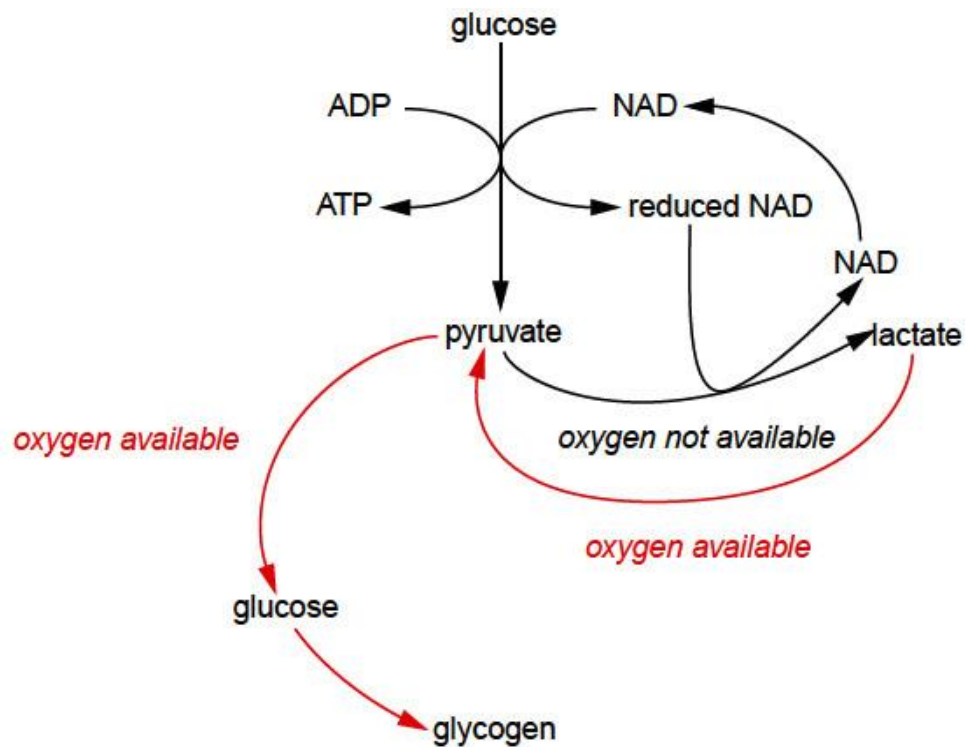


Figure 5.2

- (b) Explain why pyruvate needs to be converted to lactate in the absence of oxygen.

When oxygen is absent, cells will not be able to produce ATP by oxidative phosphorylation;
 Conversion of pyruvate to lactate is needed to regenerate NAD so that glycolysis can continue;
 NAD is needed during glycolysis to produce 2 net ATP via substrate level phosphorylation during glycolysis;
 Max2

Hexokinase is an enzyme that plays a critical role in glucose metabolism. This enzyme converts glucose into glucose-6-phosphate. This step is important as it energises glucose for further metabolic reactions. Fig. 5.3 is a computer-generated image of the enzyme hexokinase binding with its substrate, glucose.

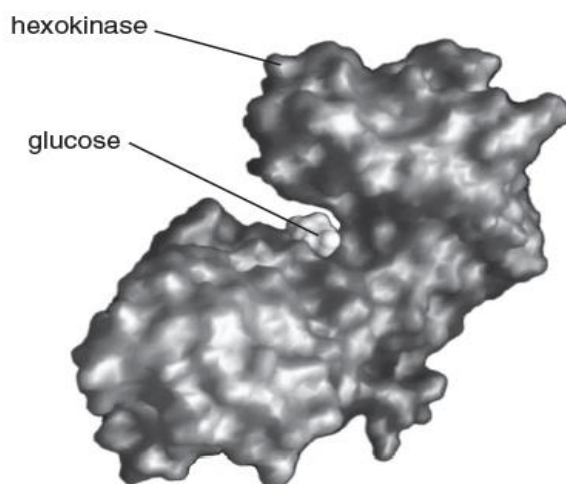


Figure 5.3

Crystallization studies have shown that the shape of hexokinase's active site is not complementary to the shape of glucose.

- (c) Describe the mechanism by which hexokinase binds to glucose.

Induced fit mechanism;
Loose association with substrate causes conformation change in enzyme's active site;
 Conformational change causes enzyme active site to be complementary to the shape of glucose;
 Allows active site to fit to substrate more precisely /allows stronger binding of substrate with enzyme's active site;
 Max 2

[2]

The phosphorylation of glucose by hexokinase also prevents it from leaving the liver cells.

- (d) Suggest why glucose-6-phosphate cannot move out of liver cells.

Glucose-6-phosphate is polar;
Hydration shell of Glucose-6-phosphate prevents it from passing through hydrophobic core of membrane;
 There are no specific transport protein for Glucose-6-phosphate to facilitate transport out of cells;
 Max 2

[2]

[Total: 11]

6. Most ATP is made in cells by membrane systems that create proton gradients by pumping protons from one compartment to another. Figure 6.1 show two such organelles.

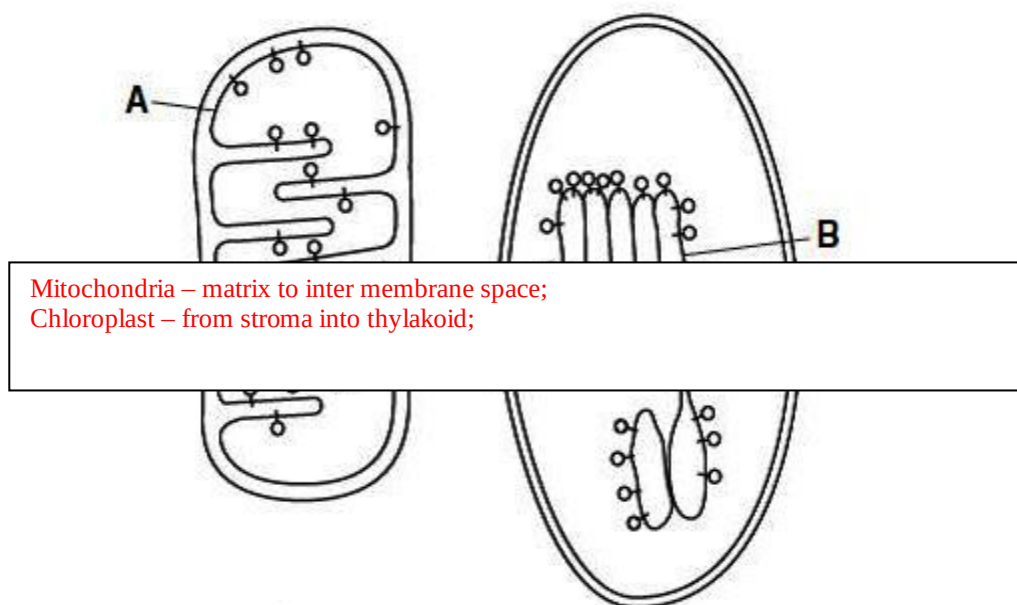


Figure 6.1

- (a) Draw arrows onto each organelle in Figure 6.1 to show the direction in which protons are pumped. [2]
- (b) In addition to facilitating the formation of proton gradients, state one other importance of compartmentalisation within organelles **A** or **B** for ATP synthesis.

Enzymes and substrates of Krebs cycle are kept in close proximity/ confined within the matrix optimising rate of reactions

OR

Enzymes and substrates of Calvin cycle are kept in close proximity/ confined within the stroma optimising rate of reactions;

Optimal conditions e.g. pH for enzymes of Krebs cycle can be maintained within matrix for higher rate of reaction;

OR

Optimal conditions e.g. pH for enzymes of Calvin cycle can be maintained within stroma for higher rate of reaction;

Prevent intermediates from different metabolic pathways from interfering from each other;

AVP

In 1882, the German botanist T.W. Engelmann performed an experiment to investigate the effects of different wavelengths of light on the rate of photosynthesis using *Spirogyra*. *Spirogyra* is a type of green alga that contains photosynthetic organelles. Other studies have shown that there are pigments that absorb blue, green, yellow and red light in these organelles.

A strand of *Spirogyra* was placed into water containing aerobic bacteria. Different parts of the strand were exposed to different colours of light. After a period of time, the bacteria had moved into the positions shown in Figure 6.2.

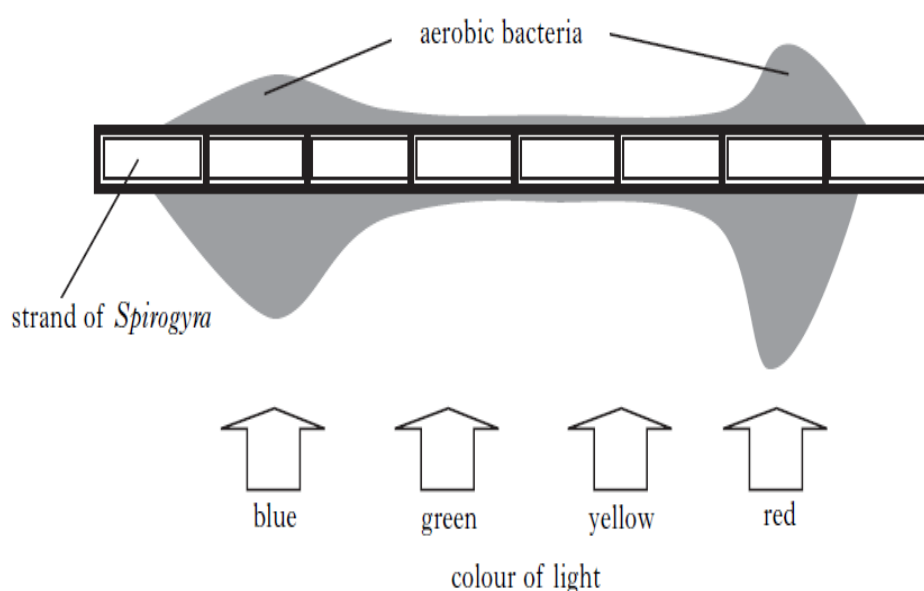


Figure 6.2

It was concluded that wavelengths corresponding to red and blue light caused a higher rate of photosynthesis compared to wavelengths corresponding to green and yellow light.

(c) With reference to Figure 6.2,

(i) explain why the abovementioned conclusion was drawn.

.....

There is a more/greater distribution of aerobic bacteria in the regions with red and blue light compared to regions with yellow and green light;

AND

Aerobic bacteria is attracted towards oxygen produced by the algae;
Oxygen is produced by photolysis of water during non-cyclic photophosphorylation within the light dependent reactions;
A greater amount of oxygen produced with red and blue light attracts more aerobic bacteria;
Max2

- (ii) suggest a role for the pigments absorbing green and yellow light in Spirogyra's photosynthetic organelles.

Broadened/widen the absorption spectrum / widen action spectrum;
They channel light energy of different wavelengths to chlorophyll a/main
photosynthetic pigment/reaction centre;
Increase the efficiency of photosynthesis;
Max 2

[Total: 8]

7. Wolf, *Canis lupus*, is one of the world's best known and well researched species, with arguably more books written about it than any other wildlife species. It is the largest extant member of its family, *Canidae*, with males averaging 45kg and females 37kg.

Wolves are further classified into a number of sub-species based on marked phenotypic differences, such as body size and colour. Figure 7.1 shows the several sub-species of wolves in existence today.

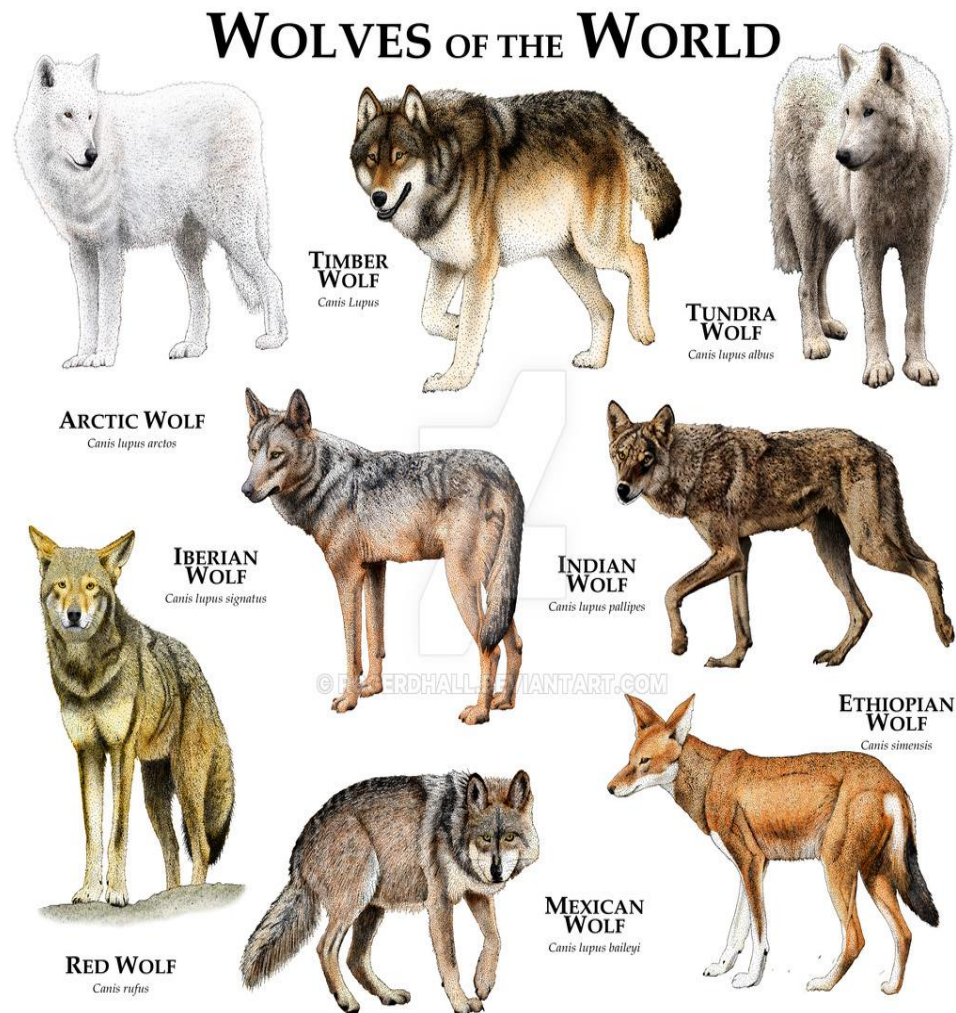


Figure 7.1

- (a) Suggest one reason why the existing populations of wolves are classified as

Can interbreed to produce fertile and viable offspring (although geographically isolated);
DNA sequence of many genes very similar;
Morphological/physiological/behavioural/biochemical features (name 2 out of the 4) are very similar but not different enough to be classified as distinct species;
Occupy same ecological niche;
Max 1

Timber wolves are native to the wilderness of Eurasia and North America. Studies have suggested that the timber wolves once co-existed with the extinct dire wolf, *Canis dirus*, on the North American continent.

Figure 7.2 depicts skeletal models of a timber and dire wolf. Dire wolves were larger and had more powerful jaws compared to timber wolves. However, they had proportionally much shorter legs than timber wolves, and were as such slower runners. They are known to feed on different prey, with the large dire wolves favouring larger prey and the timber wolves favouring smaller prey. It is known that the dire wolves went extinct approximately 8000 years ago, which coincides the time period when its large prey declined in numbers.



(b)

Using

Dire wolves faced a lack of large prey / competition from timber wolves as selection pressure;

Dire wolves had a selective disadvantage as:

Larger prey declined in numbers compared to small prey;

Timber wolves are more agile and can outcompete dire wolves for small prey;

Dire wolves were larger and slower in catching small prey;

(Max 2 marks for selective disadvantage)

AVP

Dire wolves had lower fitness compared to timber wolves and reproduced in smaller numbers eventually leading to their extinction;

Max 3

[3]

(c)

Explain why a population is the smallest unit that can evolve

Although natural selection acts on individuals, individuals do not change or evolve over their lifespan;

Variation (within the gene pool) only occurs at the population level;

Evolution is determined by change in allele frequency in a population over several generations of individuals/time.

Individuals must be able to interbreed with other members within the population to pass on alleles for any possible change in allele frequency;

Max 2

[2]

Scientists used DNA hybridisation to determine the evolutionary relationships between five members of the *Canidae* family. The temperature at which a molecule of double-stranded DNA separates into two single strands is the separation temperature. A region of non-coding DNA was analysed for the study. The scientists recorded the mean separation temperature of this region of DNA in which both strands were from the same species. The scientists then recorded the mean decrease in separation temperature of DNA in which one of the strands was from another species. Their results are shown in Table 7.1

Table 7.1

	Mean decrease in separation temperature/ °C				
	Timber wolf	Dog	Coyote	Ethiopian wolf	Dhole
Timber wolf					
Dog	1.7				
Coyote	2.3	2.3			
Ethiopian wolf	3.6	3.6	3.5		
Dhole	4.8	4.8	4.7	4.9	

- (d) With reference to Table 7.1, explain why Dholes are most distantly related to the timber wolf based on the *Canidae* members studied.

The mean decrease in separation temperature is greatest for between dhole and timber wolf at 4.8 °C as compared to other animals with respect to timber wolf;
 This implies that there are fewer hydrogen bonds/ complementary base pairs in tested DNA sequence between timber wolf and dhole;
 Fewer complementary base pairs in DNA sequences between different organisms reflect a greater number of nucleotide differences in the DNA sequence;
 Relatedness decreases with increasing number of nucleotide difference in DNA sequence (OWTTE);

The scientists assume that the decreases in separation temperatures are directly proportional to the time since the evolutionary lines of *Canidae* members separated. Dogs are thought to have separated from timber wolves 40000 years ago.

- (e) Using this information, calculate how long ago, to the nearest year, Ethiopian wolves and timber wolves separated. Show your working.

3.6/1.7 X 40,000;
84706 years;

Feature	Classification	Phylogeny
Basis of grouping	Organisms are <u>grouped based on similarities in characteristics</u> ; and does <u>not consider evolutionary relationships</u> ;	Organisms are <u>grouped into taxons</u> ; that is <u>based on evolutionary relationships</u> ;
Format of organisation	Organisation is in the <u>form of a hierarchy</u> of increasingly specific/exclusive categories;	Organisation is in the form of <u>phylogenetic tree</u> ;
Natural / man-made organisation	Classification is <u>human effort to give order to data</u> via binomial nomenclature and hierarchy to ensure that organisms can be universally identified by their scientific name;	Phylogeny reflects <u>natural relationship among organisms</u> ;

Identification vs studying relationships

:: for 1m

Marks awarded for point to point comparison

Max 2

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8.
- (a) Compare and contrast the reproductive cycles of lambda phage and HIV. [8]
 - (b) Describe the structural features and regulation of the tryptophan operon. [7]
 - (c) Explain the role of geographical and behavioural isolating mechanisms in the evolution of new species [5]
- [Total: 20]
- 9.
- (a) Compare and contrast triglyceride and starch as a storage molecule. [6]
 - (b) Explain why cancer is a multi-step disease. [8]
 - (c) Describe the role of receptor tyrosine kinases in insulin signalling. [6]
- [Total: 20]

End of Paper

(a) Compare and contrast the reproductive cycles of lambda phage and HIV. [8]

	Lambda Phage	Human immunodeficiency virus (HIV)
Host	<i>E. coli</i> ;	CD4 ⁺ T cells;
Attachment	Tail fibres bind to the surface receptor of cell	HIV GP120 on viral envelope binds to CD4 receptors on T4-lymphocytes
Penetration	Viral tail sheath contracts to injects viral DNA into the host cell.	HIV envelope fuses with host cell surface membrane to enter viral nucleocapsid containing viral genome into host cell.
Synthesis	Does not need reverse transcriptase to produce viral DNA	Uses reverse transcriptase to produce viral DNA
	When viral DNA incorporated into host chromosome, it can also leave to enter lytic cycle	Integrated viral DNA remains a permanent resident of the host cell's genome.
Release	Released by cell lysis of host cell during lytic cycle	Released by budding off from host cell
	Host cells are killed after release of virions.	Host cell still intact after reproduction. Will not die immediately.

Max 4

Both involve binding to specific receptor sites on host cell for attachment.

Both viral genomes are capable of integration into host cell's genome.

Both viruses are able to propagate without killing the host cells which they are dependent on.

Both viruses require host cell machinery for replication and synthesis of new viral particles;

Max 3

(b) Describe the structural features and regulation of the tryptophan operon. [7]

The tryptophan operon has a promoter which is the RNA polymerase binding site;
Operator which is the binding site for repressor protein;
5 structural genes which are clustered together and controlled by one promoter;
The structural genes are trpE, trpD, trpC, trpB, trpA;
These genes code for enzymes that catalyse the metabolic pathway for tryptophan synthesis;
The operator is sandwiched between the promoter and the structural genes;
Max 4

Trp operon is controlled by negative regulation
Negative regulation of lac operon involves the use of the trp repressor coded by regulatory gene trpR;
In absence of tryptophan the trp repressor is inactive and unable to bind to the operator resulting in transcription of the structural genes for tryptophan synthesis;
When tryptophan is present in high amounts, it acts as a co-repressor and binds to the trp repressor to activate it;
Activated trp repressor binds to the operator and blocks access of RNA polymerase to promoter to inhibit transcription of the trp operon;
Max 4

(c) Explain the role of geographical and behavioural isolating mechanisms in the evolution of new species [5]

Speciation occurs when there is disruption to gene flow in a population of a particular species;
Disruption of gene flow can occur by behavioural isolation;
Geographical isolation occurs when there is a physical barrier separates members of a population into 2 sub populations, preventing interbreeding;
Behavioural isolation, occurs when there is variation in behaviour in a population and members tend to mate preferentially with members with similar behaviour;
Changes in allele frequency occur independently in isolated subpopulation resulting in genetic divergence;

Genetic divergence between isolated sub-populations results in genetically distinct species which are unable to interbreed and produce fertile viable offspring;
Geographical isolation results in allopatric speciation;
Behavioural isolation results in sympatric speciation;

[Total: 20]

(a) Compare and contrast triglyceride and starch as a storage molecule. [6]

Feature	Triglyceride	Starch
Basic units	Triglyceride is made up of 1 glycerol and 3 fatty acid chains;	Starch is made up of repeating units of α glucose;
Type of Bonds	bonds linking up the monomers are ester linkages;	bonds linking up the monomers are glycosidic linkages;
Type of biomolecule	Triglyceride is not a polymer;	Starch is a polymer of glucose;
As energy store	Higher/ twice amount of energy stored/ superior long term energy store per unit mass Due to greater number of carbon atoms/ more C-H bonds for the same mass; lesser amount of fats needs to be stored for the same amount of energy;	Lower/ 2X less energy stored per unit mass; Due to more –OH groups and so less carbon atoms/ less C-H bonds for the same mass; More starch molecules need to be stored for the same amount of energy;

Max 3

Both have high percentage of hydrocarbons and hence stores energy;

Both are macromolecules formed through condensation reactions;

Both are compact;

Both are insoluble and therefore do not affect water potential of cells;

Max 3

(b) Explain why cancer is a multi-step disease.**[8]**

The development of cancer requires the accumulation of mutations;

Mutations occur in multiple genes which control regulatory checkpoints of the cell cycle;

Such mutations disrupt the normal cell cycle, thus causing the cell to undergo excessive cell growth and proliferation/division;

A gain-in-function mutation must occur in at least one proto-oncogene;

A gain-of function is a dominant mutation where mutation in just one allele of a proto-oncogene will result in oncogenic conversion;

Oncogenes promote uncontrolled cell growth and proliferation/division;

A loss-of-function must occur in at least one tumour suppressor gene;

As loss-of- function mutations is a recessive mutation, such changes must occur in both copies of the tumour suppressor gene /alleles to disrupt the function of tumour suppressor gene;

Disruption of tumour suppressor genes results in loss of ability to inhibit cell cycle progression/activate DNA repair/ activate apoptosis;

Activation/up-regulation of telomerase gene should also occur for cancer cells to divide indefinitely;

Loss of contact inhibition will enable the cells to grow into a tumour/mass of cells;

Angiogenesis must occur within the tumour so that the blood vessels formed can transport oxygen and nutrients for its growth;

Finally the cells must be able to undergo metastasis/ leave the primary site and spread to other tissues in different parts of the body via the blood stream and form multiple tumours;

(c) Describe the role of receptor tyrosine kinases in insulin signalling. [6]

Insulin as a polar peptide hormone that cannot diffuse/pass through the phospholipid bilayer of the cell membrane;

RTK spans the cell membrane/ is a transmembrane protein;

Each RTK has an extracellular insulin binding domain that is specific / has a complementary shape to that of insulin;

Upon binding of insulin to the extracellular domain, receptor polypeptides dimerise;

Each RTK contains an intracellular /cytoplasmic tyrosine kinase domain;

The tyrosine kinase domain of each polypeptide is activated upon dimerization;

Tyrosine kinase domain transfers / adds a phosphate from an ATP molecule to a tyrosine residue on the intracellular domain of the other polypeptide;

Addition of phosphate groups enables relay proteins binding to specific tyrosine residues on RTKs;

Upon binding, relay proteins undergo conformational changes that results in activation and subsequent signal transduction;

[Total: 20]